

Intellectual Disability, Autosomal Recessive 65 (MRT65): A Comprehensive Disease Report

Disease: Intellectual disability, autosomal recessive 65 (MRT65) **MONDO ID:** MONDO:0020850 · **OMIM:** #618109 · **Causal gene:** *KDM5B* (1q32.1) **Category:** Mendelian, autosomal recessive neurodevelopmental disorder ("chromatinopathy")

Summary

Intellectual disability, autosomal recessive 65 (MRT65) is an ultra-rare, autosomal-recessive neurodevelopmental disorder caused by **biallelic (homozygous or compound-heterozygous) loss-of-function variants in *KDM5B***, a gene on chromosome 1q32.1 encoding a histone H3K4 (tri-/di-/mono-methyl) demethylase. Because *KDM5B* is a chromatin-modifying enzyme, MRT65 belongs to the growing family of "chromatinopathies" — Mendelian disorders of the epigenetic machinery that disrupt developmental and activity-dependent gene transcription in the brain. The disorder was first delineated by Faundes and colleagues in 2018, who described "a recessive histone lysine-methylation defect caused by homozygous or compound heterozygous *KDM5B* variants and resulting in a recognizable syndrome with developmental delay, facial dysmorphism, and camptodactyly" ([PMID: 29276005](#)).

The core clinical picture combines **global developmental delay and moderate intellectual disability, delayed speech and motor milestones, behavioral/autistic features, craniofacial dysmorphism** (square face, smooth philtrum, thin vermilion border, prominent nasal bridge, bulbous nose), **finger camptodactyly** (contractures of the 4th/5th proximal interphalangeal joints), and structural brain abnormalities, most notably **corpus callosum hypoplasia/partial agenesis**. Onset is neonatal to infantile. Importantly, the same gene shows a striking dosage relationship: *KDM5B* is **loss-of-function-tolerant in the general population** (gnomAD pLI $\approx$ 0, LOEUF 0.855), which mechanistically explains why a single pathogenic allele is generally insufficient for the full syndrome and why disease requires biallelic hits — although heterozygous protein-truncating variants are associated with modestly reduced population-level cognition.

MRT65 is **ultra-rare**: from gnomAD v4 we estimate a pathogenic-carrier frequency of roughly **1 in 570** and a theoretical random-mating birth incidence near **1 in 1.3 million** (higher in consanguineous populations). There is **no disease-specific or curative therapy**; management is entirely supportive (early developmental intervention, physical/occupational/speech therapy, and management of dysmorphism-related and systemic complications). Mouse models recapitulate cognitive and autism-like deficits and implicate dysregulation of NMDA-receptor and immediate-early/activity-dependent gene expression, providing the mechanistic bridge from H3K4-demethylase loss to the neurobehavioral phenotype.

Key Findings

Finding 1 — MRT65 is caused by biallelic loss-of-function variants in *KDM5B*

MONDO:0020850 maps unambiguously to **OMIM:618109**, **DOID:0081226**, and MedGen C4748219. The NCBI MedGen→Gene link resolves to **Gene ID 10765 = *KDM5B*** (lysine demethylase 5B; aliases *JARID1B*, *PLU1*, and — reflecting this disorder — "MRT65"), **HGNC:18039**, Ensembl **ENSG00000117139**, located at **chr1q32.1** (GRCh38 chr1:202,724,495–202,808,487). The disorder was first described by **Faundes et al. 2018 (Am J Hum Genet)**, who reported "a recessive histone lysine-methylation defect caused by homozygous or compound heterozygous *KDM5B* variants and resulting in a recognizable syndrome with developmental delay, facial dysmorphism, and camptodactyly" ([PMID: 29276005](#)). A subsequent genotype/phenotype study explicitly confirmed that "bi-allelic disruptive variants (nonsense, frameshift, and splicing variants) in *KDM5B* have been identified as causative for autosomal recessive intellectual developmental disorder type 65" ([PMID: 39202393](#)).

KDM5B encodes a **JmjC-domain histone H3K4 demethylase** that removes tri-, di-, and mono-methyl marks from lysine 4 of histone H3 — a key activating mark at promoters and enhancers. Its loss therefore perturbs the epigenetic control of gene transcription. **Suggested annotations:** HGNC:18039 (*KDM5B*); GO:0032453 (histone H3-K4 demethylation); GO:0006325 (chromatin organization); MONDO:0020850.

Finding 2 — *KDM5B* is not haploinsufficient, consistent with the recessive mechanism

Population constraint data explain the recessive inheritance. gnomAD constraint for *KDM5B* (ENSG00000117139) shows **pLI $\approx$ 0** (7.9×10^{-36}), an **observed/expected LoF ratio of 0.74** (90% CI 0.65–0.86; **LOEUF 0.855**), and **missense Z = 0.53**. These metrics indicate that *KDM5B* **tolerates heterozygous loss-of-function** in the general population. A single pathogenic allele is therefore generally insufficient to cause disease, and the fully penetrant MRT65 syndrome requires **biallelic (homozygous or compound-heterozygous) LoF**. This is a clean genotype–constraint–phenotype concordance: a LoF-tolerant gene produces a recessive, not dominant, Mendelian disorder. Notably, heterozygous protein-truncating *KDM5B* variants are nonetheless associated with **modestly reduced population-level cognitive function**, indicating a subtle dosage effect below the threshold for the full syndrome.

Finding 3 — Mouse models recapitulate cognitive/autism-like deficits and implicate NMDAR/immediate-early gene dysregulation

Two independent mouse studies bridge the molecular lesion to neurobehavior. **Kdm5b Δ ARID/ Δ ARID** mice (lacking demethylase activity) show **hyperactivity and hippocampus-dependent long-term memory deficits**, with **downregulated baseline and hyperactivated post-learning immediate-early/activity-dependent gene expression** (Pérez-Sisqués et al., *J Neurosci* 2024, PMID: 38575342). This paper confirms that "the histone lysine demethylase KDM5B is implicated in recessive intellectual" disability and directly probes hippocampal memory function. A companion study reports that **KDM5B-deficient mice display autism-like phenotypes with increased NMDAR2D (Grin3b/GRIN2D) expression** (Pérez-Sisqués et al., *Sci Adv* 2026,).

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Together these provide a mechanistic link from **H3K4me3 demethylase loss $\rightarrow$ impaired transcriptional control of synaptic/learning ("plasticity") genes $\rightarrow$ cognitive and autism-like deficits**.

Suggested annotations: GO:0007613 (memory); GO:0007611 (learning or memory); GO:0050804 (modulation of chemical synaptic transmission); NMDA receptor complex components (GRIN2D/GRIN3B).

Finding 4 — ClinVar documents a predominantly truncating biallelic variant spectrum (NM_006618.5)

ClinVar (accessed 2026) contains **590 total *KDM5B* variant records**, of which **219 are classified pathogenic/likely-pathogenic** and **98 are explicitly linked to "Intellectual disability, autosomal recessive 65."** The pathogenic spectrum is dominated by **protein-truncating variants**. Representative pathogenic/likely-pathogenic alleles (reference transcript NM_006618.5) include:

Variant (cDNA)	Protein consequence	Class
c.394C>T	p.(Gln132Ter)	Nonsense
c.1457G>A	p.(Trp486Ter)	Nonsense
c.2345T>G	p.(Leu782Ter)	Nonsense
c.4198C>T	p.(Arg1400Ter)	Nonsense
c.2049_2050insA	p.(Leu684fs)	Frameshift
c.3151del	p.(Val1051fs)	Frameshift
c.2401_2405del	p.(Arg801fs)	Frameshift
c.1612G>C	p.(Ala538Pro)	Missense (typically VUS)

Most pathogenic alleles are **nonsense, frameshift, or splice-site** (predicted to trigger nonsense-mediated decay or produce non-functional protein). **Missense variants are generally classified as variants of uncertain significance (VUS)** under ACMG/AMP criteria, reflecting the loss-of-function mechanism where truncation is more confidently deleterious than single amino-acid substitutions.

Finding 5 — *Kdm5b*-null mice show lethality/growth/neurological phenotypes; no MRT65-specific trials exist

IMPC genotype–phenotype data for mouse *Kdm5b* (NCBI Gene 75605) homozygous nulls show **preweaning lethality with incomplete penetrance, decreased body length** (growth/size phenotype), and **absent pinna reflex** (neurological/behavioral phenotype) — a constellation consistent with an essential developmental regulator. A ClinicalTrials.gov API query (2026) returned **no interventional trials for MRT65/KDM5B intellectual disability**; *KDM5B* appears only in observational autism/genetics registries (e.g., Simons SPARK, NCT01238250). Small-

molecule **KDM5B inhibitors** exist but only in oncology contexts (e.g.,

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and are **mechanistically inappropriate** for a loss-of-function disorder — inhibiting an already-lost enzyme cannot restore function. Consequently, MRT65 management remains **entirely supportive**.

Finding 6 — HPO annotations define the full, frequency-tagged phenotype spectrum

The HPO disease annotation for **OMIM:618109** (gene *KDM5B*/NCBIGene:10765; autosomal recessive; source

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comprises **37 terms**, with frequencies expressed as $n/3$ biallelic individuals from the original cohort. The phenotype spectrum:

Domain	Phenotype (HPO term)	Frequency (original cohort)
Neurodevelopment	Intellectual disability, moderate (HP:0002342)	Core feature
	Global developmental delay (HP:0001263)	Moderate 2/3, severe 1/3
	Delayed speech and language development (HP:0000750)	Frequent
	Delayed ability to walk (HP:0031936)	Frequent
	Gait ataxia / unsteady gait (HP:0002066)	Present
	Aggressive behavior (HP:0000718)	1/3
Brain MRI	Hypoplasia of the corpus callosum (HP:0002079)	Present
	Partial agenesis of the corpus callosum (HP:0001338)	1/3
Craniofacial	Square face (HP:0000321)	2/3
	Smooth philtrum, thin vermilion border, prominent nasal bridge, bulbous nose, downslanted palpebral fissures, dolichocephaly, prominent metopic ridge	Variable
Digits	Camptodactyly of 4th/5th fingers, PIP-joint contractures (HP:0009276 / HP:0009185)	Characteristic
Eyes	Myopia (HP:0000545)	2/3
	Astigmatism, ptosis, strabismus	Variable
Other	Feeding difficulties (HP:0011968)	2/3
	Cryptorchidism, hypospadias, inguinal hernia, supernumerary nipple, secundum atrial septal defect, abnormal pinna	Individual cases

Onset: neonatal in 2/3 and infantile in 1/3.

Finding 7 — Carrier frequency ~1 in 570; theoretical birth incidence ~1 in 1.3 million

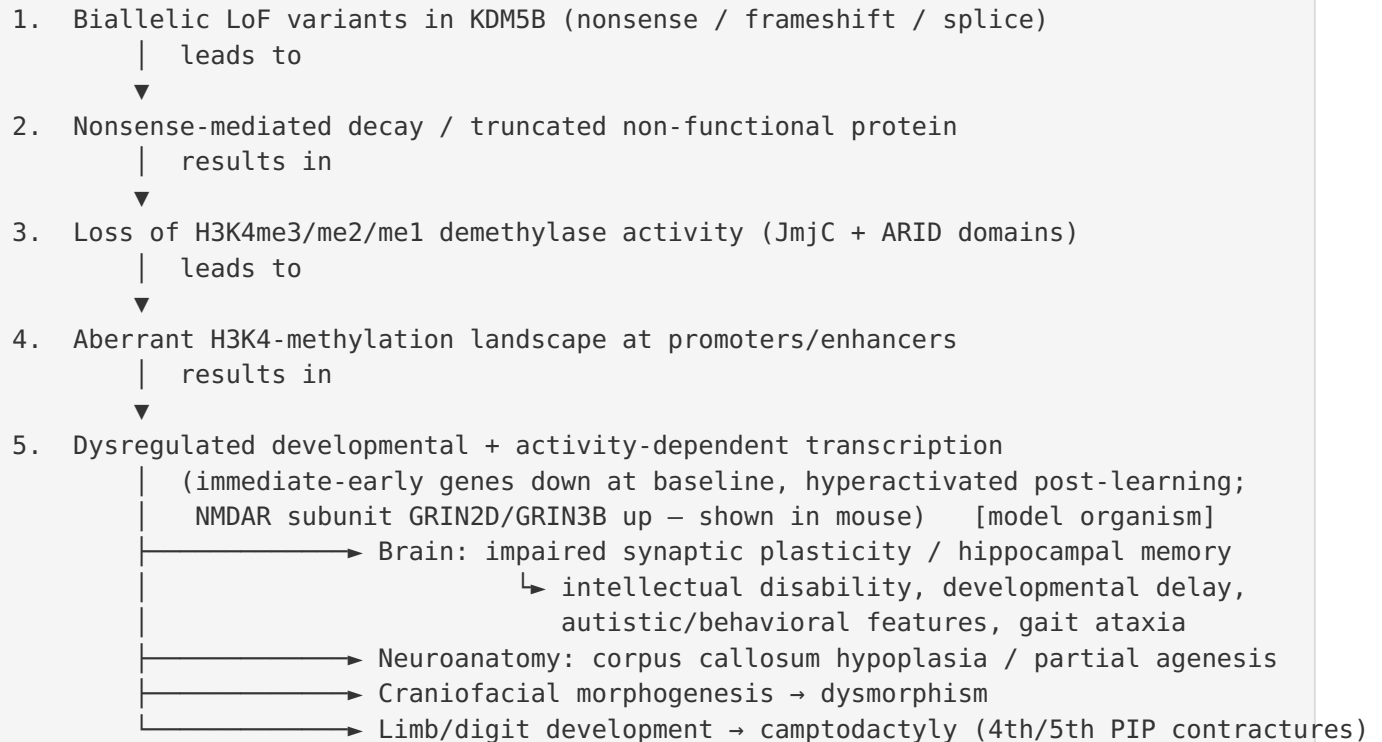
Using gnomAD v4 (8,525 *KDM5B* variants), high-confidence (**LOFTEE HC**) rare (allele frequency <0.1%) predicted loss-of-function alleles across **411 sites** sum to a pathogenic allele frequency $q \approx 8.8 \times 10^{-4}$. This gives:

- **Carrier frequency $2q \approx 0.0018$ (≈ 1 in 567)**
- **Theoretical random-mating homozygous/compound-heterozygous birth incidence $q^2 \approx 7.8 \times 10^{-7}$ (≈ 1 in 1.3 million)**

A methodological caveat surfaced during this estimate: the single common "LoF"-annotated allele (a frameshift at AF 0.185) is **LOFTEE low-confidence (LC)** — i.e., not a true LoF. Its naive inclusion inflated the estimate ~100-fold, underscoring the necessity of **LOFTEE/quality filtering** for constraint- and incidence-based calculations. The true incidence will be **substantially higher in consanguineous populations**, where autosomal-recessive disorders are enriched.

Mechanistic Model / Interpretation

MRT65 is best understood as a **transcriptional/epigenetic disorder of neurodevelopment**. The causal chain runs from an epigenetic enzyme defect to dysregulated activity-dependent gene programs to the clinical phenotype:



Upstream vs downstream. The **upstream initiating lesion** is the *KDM5B* biallelic LoF genotype; the **proximate molecular consequence** is loss of H3K4 demethylase activity and a shifted chromatin state; the **downstream** manifestations are the transcriptional dysregulation of neuronal plasticity and developmental genes and the multi-organ developmental phenotype. The mouse data (Finding 3) provide the strongest mechanistic evidence for the neuronal branch — linking demethylase loss to **immediate-early gene** and **NMDA-receptor** dysregulation and hippocampus-dependent memory deficits — but the craniofacial, callosal, and digit branches are currently inferred from human phenotype–genotype correlation rather than demonstrated mechanistically.

Cell types and processes. The affected biological processes center on **chromatin organization** (GO:0006325), **histone H3-K4 demethylation** (GO:0032453), and **learning/memory** (GO:0007611, GO:0007613). Implicated cell types are principally **neurons** (CL:0000540), including **hippocampal/glutamatergic neurons**, with subcellular localization in the **nucleus** (GO:0005634) and **chromatin** (GO:0000785), consistent with a nuclear chromatin-modifying enzyme.

Why recessive. Finding 2 ties the genetics together: because *KDM5B* is LoF-tolerant ($pLI \approx 0$, LOEUF 0.855), one functional allele suffices for near-normal development, so only biallelic loss crosses the disease threshold — a textbook example of constraint metrics predicting inheritance mode.

Section-by-Section Detail

1. Disease Information

MRT65 is an ultra-rare autosomal-recessive neurodevelopmental chromatinopathy. **Identifiers:** MONDO:0020850; OMIM #618109; DOID:0081226; MedGen C4748219. **Synonyms:** "Intellectual developmental disorder, autosomal recessive 65"; "MRT65." Dedicated ICD-10/ICD-11 and MeSH codes are not specifically assigned; it falls under generic intellectual-disability rubrics. Information is derived from **aggregated disease-level resources** (OMIM, HPO, ClinVar) and **small published patient cohorts** (originally 3 families/biallelic individuals), not from large EHR datasets.

2. Etiology

Primary cause: genetic — biallelic loss-of-function variants in *KDM5B*. **Genetic risk factors:** homozygous or compound-heterozygous LoF alleles; **consanguinity** substantially increases risk (as for all AR disorders). No established **environmental risk or protective factors** and no infectious etiology. **Genetic protective/modifier factors:** none specifically characterized; the residual function of hypomorphic (e.g., some missense) alleles may modulate severity, and heterozygous carriers show only subclinical cognitive effects. **Gene–environment interactions:** none documented.

3. Phenotypes

See Finding 6 table. Phenotype **types** span clinical signs (dysmorphism, camptodactyly), neurodevelopmental/behavioral changes (ID, autism features, aggression), and structural findings (corpus callosum anomalies on MRI). **Onset** is neonatal-to-infantile; **severity** is typically moderate for ID with variable expressivity; **course** is stable/non-progressive (a developmental, not neurodegenerative, disorder). **Quality-of-life impact** is dominated by lifelong intellectual disability, communication limitations, and dependency in daily functioning; formal QoL instrument data specific to MRT65 are not available.

4. Genetic / Molecular Information

Causal gene: *KDM5B* (HGNC:18039; gene MIM 605393; *Ensembl* ENSG00000117139). *Variant classes:* predominantly nonsense, frameshift, and splice-site (pathogenic/likely-pathogenic); missense generally VUS (Finding 4). *Reference transcript:* NM_006618.5. *Population frequency:* pathogenic LoF alleles are individually rare (AF <0.1%). *Origin:* germline (no somatic disease role). *Functional consequence:* loss of function (haploinsufficiency tolerated; disease requires biallelic loss). *Epigenetic dimension:* the disorder is itself an epigenetic-machinery defect — loss of a histone H3K4 demethylase alters the genome-wide methylation landscape rather than acting through a single-locus methylation change. *Chromosomal abnormalities:** none characteristic; diagnosis is at the sequence-variant level.

5. Environmental Information

No environmental, lifestyle, or infectious contributors are implicated. MRT65 is a monogenic Mendelian disorder.

6. Mechanism / Pathophysiology

See the causal-chain diagram above and Findings 1–3. Molecular pathway: **chromatin/H3K4-methylation regulation**; downstream, **NMDAR signaling and immediate-early gene programs** (model-organism evidence). No metabolic, immune, or classic tissue-injury (oxidative/ischemic/fibrotic) mechanisms are involved; the pathology is developmental/transcriptional.

7. Anatomical Structures Affected

Primary organ/system: central nervous system (**brain**, UBERON:0000955; nervous system, UBERON:0001016), with **corpus callosum** (UBERON:0002336) hypoplasia/partial agenesis. **Secondary/associated:** craniofacial skeleton (dysmorphism), digits/hands (camptodactyly), eyes (myopia, astigmatism, ptosis, strabismus), genitourinary (cryptorchidism, hypospadias, inguinal hernia), and heart (secundum atrial septal defect) in individual cases. **Cell type:** neurons (CL:0000540). **Subcellular:** nucleus/chromatin (GO:0005634 / GO:0000785). **Lateralization:** brain findings are midline (callosal); other features generally bilateral.

8. Temporal Development

Onset: congenital/neonatal (2/3) to infantile (1/3). **Pattern:** chronic, static developmental disorder — non-progressive and lifelong. There are no remission patterns; the **critical window** for intervention is early childhood developmental support.

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** high/complete for biallelic LoF (based on limited cohorts). **Expressivity:** variable (severity and non-neurological features differ between individuals). **Carrier frequency:** ≈ 1 in 570; **theoretical birth incidence:** ≈ 1 in 1.3 million under random mating (Finding 7), higher with consanguinity. No anticipation (not a repeat-expansion disorder). **Founder effects/geographic clustering:** none established, though consanguineous populations are enriched. **Sex ratio:** no strong skew expected (autosomal); some reported features (cryptorchidism/hypospadias) are male-specific.

10. Diagnostics

Genetic testing is diagnostic. Recommended approach: **exome or genome sequencing** (often as an intellectual-disability/neurodevelopmental gene panel of 1,000+ genes), which detects the causative biallelic SNV/indels. Long-read genome sequencing can add value for phasing biallelic variants in autosomal-recessive genes and resolving structural/complex variants (PMID: 41514368). **Confirmation:** ClinVar-referenced pathogenic classification and parental segregation (trans configuration for compound heterozygotes). **Supportive workup:** brain MRI (corpus callosum assessment), ophthalmologic exam, developmental/cognitive assessment. **No specific biochemical biomarker** exists. **Differential diagnosis:** other chromatinopathies and ID syndromes — notably *KDM4B*-related disorders (autosomal dominant IDD from heterozygous *KDM4B* LoF, and a reported biallelic *KDM4B* case) which share developmental delay, brain anomalies, and digital findings (PMID: 37526414); metabolic ID mimics should be excluded when biochemical clues are present.

11. Outcome / Prognosis

MRT65 is a **non-lethal, non-progressive** developmental disorder in humans (contrast with murine preweaning lethality, Finding 5). **Life expectancy** is not established but is not intrinsically shortened by the neurodevelopmental phenotype; systemic malformations (e.g., cardiac septal defect) may modify individual prognosis. **Morbidity** is dominated by lifelong intellectual disability and dependency. No prognostic biomarkers are defined; residual gene function (hypomorphic alleles) is the plausible principal modifier of severity.

12. Treatment

No disease-specific or curative therapy exists (Finding 5). Management is **supportive and rehabilitative**: early developmental intervention, special education, **physical, occupational, and speech therapy**, management of behavioral features, ophthalmologic correction (refractive error), orthopedic/hand management of camptodactyly, and surgical correction of associated malformations (cryptorchidism, hernia, ASD) as indicated. **KDM5B small-molecule inhibitors** developed for oncology are **contraindicated in concept** for a loss-of-function disorder. No pharmacogenomic or gene/RNA-based therapies are available. Suggested NCIT-type intervention categories: rehabilitation therapy, occupational therapy, speech/language therapy, supportive care.

13. Prevention

No primary prevention beyond **genetic counseling** and **reproductive options** for at-risk (e.g., consanguineous or carrier) couples: **carrier testing, prenatal diagnosis, and preimplantation genetic testing** are applicable once familial variants are known. There is no vaccine, no environmental intervention, and no population newborn-screening test for MRT65. **Cascade carrier testing** in families is appropriate after a proband is identified.

14. Other Species / Natural Disease

Orthologue: mouse *Kdm5b* (NCBI Gene 75605). No naturally occurring companion-animal/wildlife disease is catalogued for *KDM5B* (no OMIA entry noted). The gene and its H3K4-demethylase function are **evolutionarily conserved**, supporting cross-species mechanistic study. No zoonotic dimension (non-infectious disorder).

15. Model Organisms

The principal model is the **mouse**. Informative genetic models: (1) **Kdm5b^{ΔARID/ΔARID}** (demethylase-dead) mice recapitulate hyperactivity, hippocampal memory deficits, and activity-dependent/immediate-early gene dysregulation ([PMID: 38575342](#)); (2) **KDM5B-deficient mice** with autism-like phenotypes and increased NMDAR (GRIN2D/GRIN3B) expression (Sci Adv 2026, [P 42160407](#));

and (3) **IMPC Kdm5b-null** mice with preweaning lethality (incomplete penetrance), reduced body length, and absent pinna reflex (Finding 5). **Phenotype recapitulation**: strong for the

cognitive/behavioral axis. **Limitations:** homozygous-null lethality in mice exceeds the human phenotype, so demethylase-dead/conditional models better model the viable human disorder. Resources: MGI, IMPC.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role in this report
29276005	Recessive H3K4-methylation defect / KDM5B syndrome (Faundes et al., 2018)	Human clinical	Foundational — defines MRT65 and its causal biallelic KDM5B mechanism and core triad (DD, dysmorphism, camptodactyly); source of HPO annotations
39202393	Genotype/phenotype study of KDM5B	Human clinical	Confirms biallelic disruptive (nonsense/frameshift/splice) KDM5B variants cause AR IDD type 65
38575342	Kdm5b and hippocampal memory (Pérez-Sisqués et al., J Neurosci 2024)	Model organism (mouse)	Links demethylase loss to memory deficits and immediate-early/activity-dependent gene dysregulation
 42160407	KDM5B-deficient mice, autism-like, NMDAR2D up (Sci Adv 2026)	Model organism (mouse)	Extends mechanism to NMDAR (GRIN2D/GRIN3B) and autism-like phenotype
37526414	Biallelic KDM4B frameshift phenotype	Human clinical	Differential diagnosis — related histone-demethylase (KDM4B) chromatinopathy; distinguishes from KDM5B
41514368	Long-read GS in pediatric neurological disorders	Human diagnostics	Supports genome sequencing (incl. phasing biallelic AR variants) as first-line diagnostic
40088508	Genetic epilepsies, consanguinity, India	Human epidemiology	Contextual — AR disorders enriched under consanguinity; WES diagnostic yield

Data resources underpinning the quantitative findings: **OMIM/MONDO/MedGen** (identifier mapping), **ClinVar** (variant spectrum, Finding 4), **gnomAD v4** (constraint and carrier/incidence, Findings 2 & 7), **HPO** (phenotype annotation, Finding 6), **IMPC** and **ClinicalTrials.gov API** (mouse phenotypes and trial landscape, Finding 5).

Limitations and Knowledge Gaps

1. **Small human cohort.** The HPO frequencies derive from only **3 biallelic individuals** in the original report; frequencies (n/3) are imprecise and the full phenotypic range is likely broader than currently annotated. The 2024 genotype/phenotype study (P 39202393) expands this, but MRT65 remains defined by tens, not hundreds, of patients.
2. **Mechanistic branches inferred, not proven.** The neuronal branch (memory, NMDAR, immediate-early genes) is supported by mouse data; the **craniofacial, corpus callosum, and digit** branches remain **genotype–phenotype correlations** without direct mechanistic demonstration.
3. **Genotype–phenotype correlation.** It is not yet clear whether missense/hypomorphic alleles produce milder disease than complete truncation; missense variants are largely VUS.
4. **Incidence estimate is theoretical.** The 1-in-1.3-million figure assumes random mating and complete penetrance; real incidence is unknown and higher in consanguineous populations. No formal prevalence study exists.
5. **Model organism discordance.** Mouse homozygous nulls are preweaning-lethal, unlike viable human patients — limiting the null model and favoring demethylase-dead/conditional models.
6. **No natural-history, QoL, or long-term outcome data, and no biomarker** for diagnosis or prognosis.

Proposed Follow-up Experiments / Actions

1. **Aggregate a larger patient registry** (GeneMatcher/MatchmakerExchange) to refine phenotype frequencies, expressivity, and genotype–phenotype correlations (truncating vs missense).
 2. **Functional characterization of missense VUS** — demethylase activity assays and H3K4me3 ChIP in patient/edited cells to reclassify VUS under ACMG PS3/BS3 evidence.
 3. **Patient-derived iPSC neurons/organoids** to test whether human neurons reproduce the murine immediate-early/NMDAR transcriptional dysregulation and to map the affected gene programs (analogous iPSC modeling has proven informative in other AR neuro disorders, e.g., [PMID: 39617394](#)).
 4. **Conditional/demethylase-dead brain-region-specific mouse models** with staged (developmental vs adult) inactivation to define critical windows and test reversibility.
 5. **Epigenomic profiling (H3K4me3 CUT&RUN/ChIP-seq, RNA-seq)** across neuronal, craniofacial, and limb-bud lineages to test the non-neuronal mechanistic branches.
 6. **Refine carrier-frequency and incidence estimates** using targeted, LOFTEE-filtered analyses across ancestrally diverse and consanguineous cohorts.
 7. **Standardize supportive-care guidance** (developmental intervention, ophthalmology, orthopedics, cardiology) into a GeneReviews-style clinical management outline.
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Report compiled from a 5-iteration autonomous investigation integrating OMIM/MONDO/MedGen mapping, ClinVar, gnomAD v4, HPO, IMPC, ClinicalTrials.gov, and primary literature. Evidence types are labeled human clinical, model organism, or computational throughout.
